research

June 1, 2026

[1]:
```
!cd ~/c/iu9-sql-compiler/ && ./build-release/bin/benchmarks␣
↪--benchmark_filter='^OperatorCost/' --benchmark_out=/tmp/operator-cost.json␣
↪--benchmark_out_format=json
```

```
2026-05-31T16:14:53+03:00
Running ./build-release/bin/benchmarks
Run on (8 X 4200 MHz CPU s)
CPU Caches:
  L1 Data 48 KiB (x4)
  L1 Instruction 32 KiB (x4)
  L2 Unified 1280 KiB (x4)
  L3 Unified 8192 KiB (x1)
Load Average: 4.44, 2.58, 1.61
***WARNING*** CPU scaling is enabled, the benchmark real time measurements may
be noisy and will incur extra overhead.
-------------------------------------------------------------------------------
-------------------------------
Benchmark                                                      Time
CPU   Iterations UserCounters…
-------------------------------------------------------------------------------
-------------------------------
OperatorCost/SeqScan/1024/real_time                        79997
ns        78049 ns         8741 model_cost=102.4k
output_rows=1.024k plan_cost=102.4k rows=1.024k
OperatorCost/SeqScan/2048/real_time
144761 ns       142224 ns         5232 model_cost=204.8k
output_rows=2.048k plan_cost=204.8k rows=2.048k
OperatorCost/SeqScan/4096/real_time
338605 ns       327024 ns         2056 model_cost=409.6k
output_rows=4.096k plan_cost=409.6k rows=4.096k
OperatorCost/SeqScan/8192/real_time
813515 ns       787366 ns          726 model_cost=819.2k
output_rows=8.192k plan_cost=819.2k rows=8.192k
OperatorCost/SeqScan/16384/real_time
2055770 ns      1963689 ns          575 model_cost=1.6384M
```

```
output_rows=16.384k plan_cost=1.6384M rows=16.384k
OperatorCost/SeqScan/32768/real_time

3635850 ns      3580957 ns          159 model_cost=3.2768M
output_rows=32.768k plan_cost=3.2768M rows=32.768k
OperatorCost/SeqScan/65536/real_time

7848736 ns      7709842 ns          101 model_cost=6.5536M
output_rows=65.536k plan_cost=6.5536M rows=65.536k
OperatorCost/SeqScan/131072/real_time

14549342 ns     14285211 ns           48 model_cost=13.1072M
output_rows=131.072k plan_cost=13.1072M rows=131.072k
OperatorCost/SeqScan/262144/real_time

46805723 ns     44962657 ns           24 model_cost=26.2144M
output_rows=262.144k plan_cost=26.2144M rows=262.144k
OperatorCost/Filter/1024/real_time

110848 ns       109579 ns         6051 model_cost=102.4k
output_rows=512 plan_cost=204.8k rows=1.024k
OperatorCost/Filter/2048/real_time

175851 ns       174124 ns         4178 model_cost=204.8k
output_rows=1.024k plan_cost=409.6k rows=2.048k
OperatorCost/Filter/4096/real_time

315093 ns       312199 ns         2166 model_cost=409.6k
output_rows=2.048k plan_cost=819.2k rows=4.096k
OperatorCost/Filter/8192/real_time

673538 ns       667479 ns         1078 model_cost=819.2k
output_rows=4.096k plan_cost=1.6384M rows=8.192k
OperatorCost/Filter/16384/real_time

1303801 ns      1294736 ns          456 model_cost=1.6384M
output_rows=8.192k plan_cost=3.2768M rows=16.384k
OperatorCost/Filter/32768/real_time

2952031 ns      2928449 ns          256 model_cost=3.2768M
output_rows=16.384k plan_cost=6.5536M rows=32.768k
OperatorCost/Filter/65536/real_time

7472369 ns      7365271 ns          102 model_cost=6.5536M
output_rows=32.768k plan_cost=13.1072M rows=65.536k
OperatorCost/Filter/131072/real_time

14015431 ns     13847255 ns           58 model_cost=13.1072M
output_rows=65.536k plan_cost=26.2144M rows=131.072k
OperatorCost/Filter/262144/real_time

27073700 ns     26782809 ns           24 model_cost=26.2144M
output_rows=131.072k plan_cost=52.4288M rows=262.144k
```

```
OperatorCost/Projection/1024/real_time
131940 ns       130699 ns         5179 model_cost=22.528k
output_rows=1.024k plan_cost=124.928k rows=1.024k
OperatorCost/Projection/2048/real_time
226190 ns       225141 ns         2989 model_cost=45.056k
output_rows=2.048k plan_cost=249.856k rows=2.048k
OperatorCost/Projection/4096/real_time
514393 ns       509783 ns         1536 model_cost=90.112k
output_rows=4.096k plan_cost=499.712k rows=4.096k
OperatorCost/Projection/8192/real_time
945676 ns       940030 ns          718 model_cost=180.224k
output_rows=8.192k plan_cost=0.999424M rows=8.192k
OperatorCost/Projection/16384/real_time
1904450 ns      1892580 ns          366 model_cost=360.448k
output_rows=16.384k plan_cost=1.99885M rows=16.384k
OperatorCost/Projection/32768/real_time
3678534 ns      3656980 ns          197 model_cost=720.896k
output_rows=32.768k plan_cost=3.9977M rows=32.768k
OperatorCost/Projection/65536/real_time
7472132 ns      7428647 ns           93 model_cost=1.44179M
output_rows=65.536k plan_cost=7.99539M rows=65.536k
OperatorCost/Projection/131072/real_time
15381800 ns     15295836 ns           46 model_cost=2.88358M
output_rows=131.072k plan_cost=15.9908M rows=131.072k
OperatorCost/Projection/262144/real_time
41055491 ns     40717845 ns           19 model_cost=5.76717M
output_rows=262.144k plan_cost=31.9816M rows=262.144k
OperatorCost/Sort/1024/real_time
162085 ns       160916 ns         4340 model_cost=123.904k
output_rows=1.024k plan_cost=226.304k rows=1.024k
OperatorCost/Sort/2048/real_time
346371 ns       344873 ns         2062 model_cost=270.336k
output_rows=2.048k plan_cost=475.136k rows=2.048k
OperatorCost/Sort/4096/real_time
800255 ns       792542 ns          883 model_cost=585.728k
output_rows=4.096k plan_cost=995.328k rows=4.096k
OperatorCost/Sort/8192/real_time
1840524 ns      1830438 ns          379 model_cost=1.26157M
output_rows=8.192k plan_cost=2.08077M rows=8.192k
```

```
OperatorCost/Sort/16384/real_time
4673407 ns      4626060 ns          164 model_cost=2.70336M
output_rows=16.384k plan_cost=4.34176M rows=16.384k
OperatorCost/Sort/32768/real_time
10560526 ns     10441042 ns           60 model_cost=5.76717M
output_rows=32.768k plan_cost=9.04397M rows=32.768k
OperatorCost/Sort/65536/real_time
27512323 ns     27313637 ns           26 model_cost=12.2552M
output_rows=65.536k plan_cost=18.8088M rows=65.536k
OperatorCost/Sort/131072/real_time
76878972 ns     76256341 ns            9 model_cost=25.9523M
output_rows=131.072k plan_cost=39.0595M rows=131.072k
OperatorCost/Sort/262144/real_time
210729772 ns    208907082 ns            3 model_cost=54.7881M
output_rows=262.144k plan_cost=81.0025M rows=262.144k
OperatorCost/Aggregation/1024/real_time
367198 ns       365161 ns         2007 model_cost=522.24k
output_rows=1.024k plan_cost=624.64k rows=1.024k
OperatorCost/Aggregation/2048/real_time
714949 ns       711449 ns         1030 model_cost=1.04448M
output_rows=2.048k plan_cost=1.24928M rows=2.048k
OperatorCost/Aggregation/4096/real_time
1378878 ns      1372416 ns          506 model_cost=2.08896M
output_rows=4.096k plan_cost=2.49856M rows=4.096k
OperatorCost/Aggregation/8192/real_time
2970154 ns      2953446 ns          233 model_cost=4.17792M
output_rows=8.192k plan_cost=4.99712M rows=8.192k
OperatorCost/Aggregation/16384/real_time
7566779 ns      7487774 ns           94 model_cost=8.35584M
output_rows=16.384k plan_cost=9.99424M rows=16.384k
OperatorCost/Aggregation/32768/real_time
21294147 ns     21136412 ns           28 model_cost=16.7117M
output_rows=32.768k plan_cost=19.9885M rows=32.768k
OperatorCost/Aggregation/65536/real_time
58811825 ns     58334161 ns           11 model_cost=33.4234M
output_rows=65.536k plan_cost=39.977M rows=65.536k
OperatorCost/Aggregation/131072/real_time
145317421 ns    144445410 ns            5 model_cost=66.8467M
output_rows=131.072k plan_cost=79.9539M rows=131.072k
```

```
OperatorCost/Aggregation/262144/real_time
341980080 ns    339244361 ns            2 model_cost=133.693M
output_rows=262.144k plan_cost=159.908M rows=262.144k
OperatorCost/HashJoin/1024/1024/real_time
203598 ns       202661 ns         3434 lhs_rows=1.024k
model_cost=141.312k output_rows=1.024k plan_cost=346.112k
rhs_rows=1.024k
OperatorCost/HashJoin/2048/2048/real_time
383152 ns       381209 ns         1850 lhs_rows=2.048k
model_cost=282.624k output_rows=2.048k plan_cost=692.224k
rhs_rows=2.048k
OperatorCost/HashJoin/4096/4096/real_time
762752 ns       757843 ns          903 lhs_rows=4.096k
model_cost=565.248k output_rows=4.096k plan_cost=1.38445M
rhs_rows=4.096k
OperatorCost/HashJoin/8192/8192/real_time
1662913 ns      1651698 ns          461 lhs_rows=8.192k
model_cost=1.1305M output_rows=8.192k plan_cost=2.7689M
rhs_rows=8.192k
OperatorCost/HashJoin/16384/16384/real_time
4006295 ns      3972214 ns          191 lhs_rows=16.384k
model_cost=2.26099M output_rows=16.384k plan_cost=5.53779M
rhs_rows=16.384k
OperatorCost/HashJoin/32768/32768/real_time
8204329 ns      8095465 ns           82 lhs_rows=32.768k
model_cost=4.52198M output_rows=32.768k plan_cost=11.0756M
rhs_rows=32.768k
OperatorCost/HashJoin/65536/65536/real_time
16943075 ns     16813719 ns           43 lhs_rows=65.536k
model_cost=9.04397M output_rows=65.536k plan_cost=22.1512M
rhs_rows=65.536k
OperatorCost/NestedLoopJoin/64/64/real_time
457941 ns       454351 ns         1627 lhs_rows=64
model_cost=286.72k output_rows=64 plan_cost=299.52k rhs_rows=64
OperatorCost/NestedLoopJoin/128/128/real_time
1326260 ns      1316630 ns          462 lhs_rows=128
model_cost=1.14688M output_rows=128 plan_cost=1.17248M rhs_rows=128
OperatorCost/NestedLoopJoin/256/256/real_time
4755416 ns      4716712 ns          148 lhs_rows=256
model_cost=4.58752M output_rows=256 plan_cost=4.63872M rhs_rows=256
```

```
OperatorCost/NestedLoopJoin/512/512/real_time
18824481 ns     18740105 ns           39 lhs_rows=512
model_cost=18.3501M output_rows=512 plan_cost=18.4525M rhs_rows=512
OperatorCost/NestedLoopJoin/1024/1024/real_time
69821057 ns     69534977 ns           10 lhs_rows=1.024k
model_cost=73.4003M output_rows=1.024k plan_cost=73.6051M
rhs_rows=1.024k
OperatorCost/NestedLoopJoin/2048/2048/real_time
275065905 ns    274101250 ns            3 lhs_rows=2.048k
model_cost=293.601M output_rows=2.048k plan_cost=294.011M
rhs_rows=2.048k
OperatorCost/NestedLoopCrossJoin/32/32/real_time
161658 ns       160319 ns         4263 lhs_rows=32
model_cost=106.496k output_rows=1.024k plan_cost=112.896k
rhs_rows=32
OperatorCost/NestedLoopCrossJoin/64/64/real_time
391432 ns       386106 ns         1896 lhs_rows=64
model_cost=425.984k output_rows=4.096k plan_cost=438.784k
rhs_rows=64
OperatorCost/NestedLoopCrossJoin/128/128/real_time
1644326 ns      1610732 ns          331 lhs_rows=128
model_cost=1.70394M output_rows=16.384k plan_cost=1.72954M
rhs_rows=128
OperatorCost/NestedLoopCrossJoin/256/256/real_time
6994495 ns      6864530 ns           87 lhs_rows=256
model_cost=6.81574M output_rows=65.536k plan_cost=6.86694M
rhs_rows=256
OperatorCost/NestedLoopCrossJoin/512/512/real_time
27484957 ns     27127956 ns           25 lhs_rows=512
model_cost=27.263M output_rows=262.144k plan_cost=27.3654M
rhs_rows=512
OperatorCost/NestedLoopCrossJoin/1024/1024/real_time
130720901 ns    129132758 ns            5 lhs_rows=1.024k
model_cost=109.052M output_rows=1.04858M plan_cost=109.257M
rhs_rows=1.024k
```

[2]:
```
!ls -l /tmp | grep operator-cost
```

```
-rw-r--r--  1 st     users   40311 May 31 16:16 operator-cost.json
-rw-r--r--  1 st     users   33223 May 31 16:03 operator-cost-matched-
calibrated-smoke.json
-rw-r--r--  1 st     users   33307 May 31 16:13 operator-cost-matched.json
```

```
-rw-r--r--  1 st     users    6149 May 31 15:08 operator-cost-matched-plan-cost-
smoke.json
-rw-r--r--  1 st     users    2010 May 30 19:02 operator-cost-smoke.json
```

[ ]:
```
import json

results = None
with open('/tmp/operator-cost.json') as f:
    results = json.load(f)

results
```

[4]:
```
import pandas as pd

assert results

df = pd.DataFrame(results['benchmarks'])
df = df[df['run_type'] == 'iteration'].copy()

parts = df['name'].str.split('/')
df['operator'] = parts.str[1]
df['left_rows'] = parts.str[2].astype(int)
df['right_rows'] = parts.apply(lambda p: int(p[3]) if len(p) == 5 else None)

df['real_time_ms'] = df['real_time'] / 1e6

df[['operator', 'left_rows', 'right_rows', 'real_time_ms', 'model_cost']].
 ↪sort_values(['operator', 'left_rows'])
```

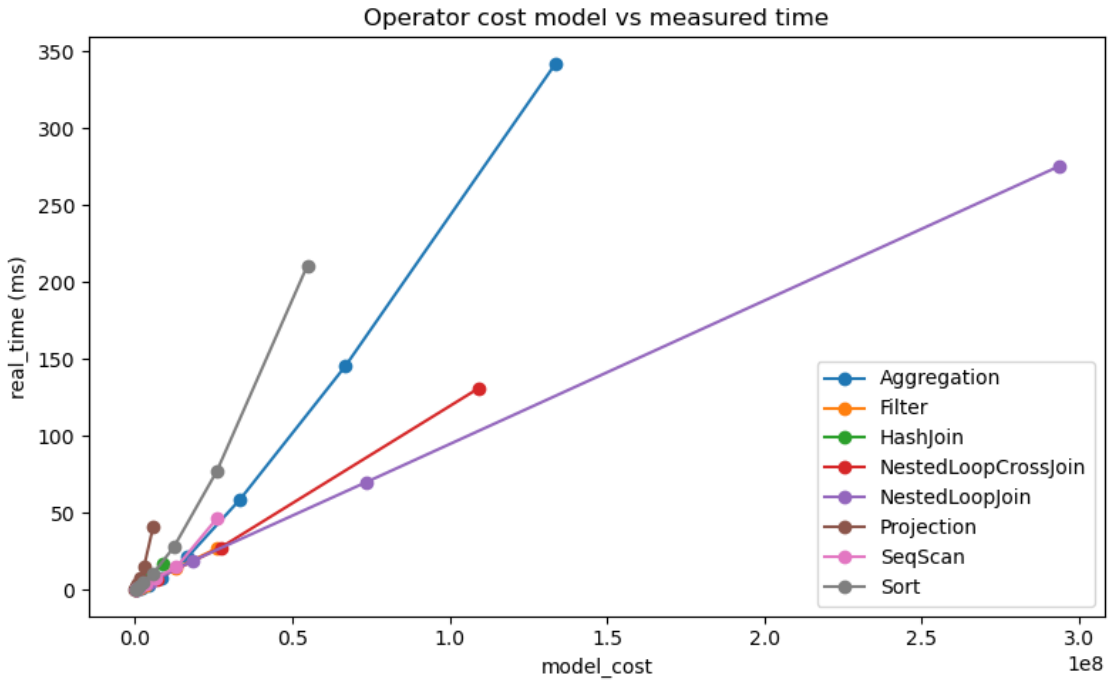
[4]:

	operator	left_rows	right_rows	real_time_ms	model_cost
36	Aggregation	1024	NaN	0.367198	522240.0
37	Aggregation	2048	NaN	0.714949	1044480.0
38	Aggregation	4096	NaN	1.378878	2088960.0
39	Aggregation	8192	NaN	2.970154	4177920.0
40	Aggregation	16384	NaN	7.566779	8355840.0
..	…	…	…	…	…
31	Sort	16384	NaN	4.673407	2703360.0
32	Sort	32768	NaN	10.560526	5767168.0
33	Sort	65536	NaN	27.512323	12255232.0
34	Sort	131072	NaN	76.878972	25952256.0
35	Sort	262144	NaN	210.729772	54788096.0

[64 rows x 5 columns]

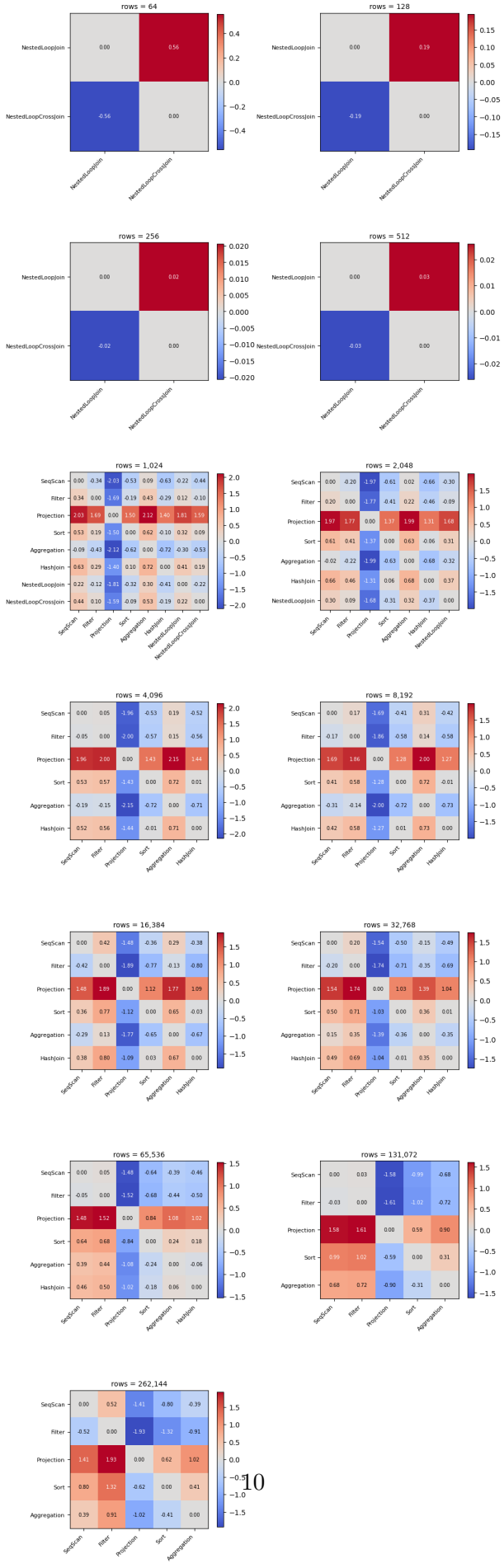
[5]:
```
import matplotlib.pyplot as plt

fig, ax = plt.subplots(figsize=(8, 5))
for op, grp in df.groupby('operator'):
```

```
    ax.plot(grp['model_cost'], grp['real_time_ms'], marker='o', label=op)

ax.set_xlabel('model_cost')
ax.set_ylabel('real_time (ms)')
ax.set_title('Operator cost model vs measured time')
ax.legend()
plt.tight_layout()
plt.show()
```



[6]:
```
import matplotlib.pyplot as plt
import numpy as np

pivot = df.set_index(['operator', 'left_rows'])[['cpu_time', 'model_cost']]

all_counts = sorted(df['left_rows'].unique())
valid_counts = [rc for rc in all_counts
                if pivot.index.get_level_values('left_rows').isin([rc]).sum()␣
 ↪>= 2]

ncols = 2
nrows = (len(valid_counts) + ncols - 1) // ncols
fig, axes = plt.subplots(nrows, ncols, figsize=(5 * ncols, 4.5 * nrows),␣
 ↪squeeze=False)
```

```
for idx, rc in enumerate(valid_counts):
    ax = axes[idx // ncols][idx % ncols]
    sub = pivot.xs(rc, level='left_rows')
    ops = sub.index.tolist()
    m = len(ops)

    mat = np.zeros((m, m))
    for i, a in enumerate(ops):
        for j, b in enumerate(ops):
            if i != j:
                mat[i, j] = (np.log(sub.loc[a, 'cpu_time'])   - np.log(sub.
 ↪loc[b, 'cpu_time']) -
                             (np.log(sub.loc[a, 'model_cost']) - np.log(sub.
 ↪loc[b, 'model_cost'])))

    vmax = np.abs(mat).max() or 1.0
    im = ax.imshow(mat, cmap='coolwarm', vmin=-vmax, vmax=vmax)
    ax.set_xticks(range(m)); ax.set_xticklabels(ops, rotation=45, ha='right',␣
 ↪fontsize=8)
    ax.set_yticks(range(m)); ax.set_yticklabels(ops, fontsize=8)
    ax.set_title(f'rows = {rc:,}', fontsize=10)
    plt.colorbar(im, ax=ax, shrink=0.8)
    for i in range(m):
        for j in range(m):
            ax.text(j, i, f'{mat[i,j]:.2f}', ha='center', va='center',␣
 ↪fontsize=7,
                    color='black' if abs(mat[i,j]) < vmax * 0.6 else 'white')

for idx in range(len(valid_counts), nrows * ncols):
    axes[idx // ncols][idx % ncols].set_visible(False)

plt.tight_layout()
plt.show()
```


rows = 64
NestedLoopJoin
NestedLoopCrossJoin
0.00
0.56
-0.56
0.00
rows = 128
0.00
0.19
-0.19
0.00
rows = 256
0.00
0.02
-0.02
0.00
rows = 512
0.00
0.03
-0.03
0.00
rows = 1,024
SeqScan
Filter
Projection
Sort
Aggregation
HashJoin
NestedLoopJoin
NestedLoopCrossJoin
rows = 2,048
rows = 4,096
rows = 8,192
rows = 16,384
rows = 32,768
rows = 65,536
rows = 131,072
rows = 262,144

[7]:
```
rows = []
for rc in valid_counts:
    sub = pivot.xs(rc, level='left_rows')
    ops = sub.index.tolist()
    for a in ops:
        for b in ops:
            if a != b:
                delta = (np.log(sub.loc[a, 'cpu_time'])   - np.log(sub.loc[b,␣
 ↪'cpu_time']) -
                         (np.log(sub.loc[a, 'model_cost']) - np.log(sub.loc[b,␣
 ↪'model_cost'])))
                rows.append({'row_count': rc, 'a': a, 'b': b, 'delta': delta})

comparison = pd.DataFrame(rows)
comparison.pivot_table(index=['a', 'b'], columns='row_count', values='delta').
 ↪round(3)
```

[7]:

row_count		64	128	256	512
a	b				
Aggregation	Filter	NaN	NaN	NaN	NaN
	HashJoin	NaN	NaN	NaN	NaN
	NestedLoopCrossJoin	NaN	NaN	NaN	NaN
	NestedLoopJoin	NaN	NaN	NaN	NaN
	Projection	NaN	NaN	NaN	NaN
	SeqScan	NaN	NaN	NaN	NaN
	Sort	NaN	NaN	NaN	NaN
Filter	Aggregation	NaN	NaN	NaN	NaN
	HashJoin	NaN	NaN	NaN	NaN
	NestedLoopCrossJoin	NaN	NaN	NaN	NaN
	NestedLoopJoin	NaN	NaN	NaN	NaN
	Projection	NaN	NaN	NaN	NaN
	SeqScan	NaN	NaN	NaN	NaN
	Sort	NaN	NaN	NaN	NaN
HashJoin	Aggregation	NaN	NaN	NaN	NaN
	Filter	NaN	NaN	NaN	NaN
	NestedLoopCrossJoin	NaN	NaN	NaN	NaN
	NestedLoopJoin	NaN	NaN	NaN	NaN
	Projection	NaN	NaN	NaN	NaN
	SeqScan	NaN	NaN	NaN	NaN
	Sort	NaN	NaN	NaN	NaN
NestedLoopCrossJoin	Aggregation	NaN	NaN	NaN	NaN
	Filter	NaN	NaN	NaN	NaN
	HashJoin	NaN	NaN	NaN	NaN
	NestedLoopJoin	-0.559	-0.194	-0.021	-0.026

	Projection	NaN	NaN	NaN	NaN
	SeqScan	NaN	NaN	NaN	NaN
	Sort	NaN	NaN	NaN	NaN
NestedLoopJoin	Aggregation	NaN	NaN	NaN	NaN
	Filter	NaN	NaN	NaN	NaN
	HashJoin	NaN	NaN	NaN	NaN
	NestedLoopCrossJoin	0.559	0.194	0.021	0.026
	Projection	NaN	NaN	NaN	NaN
	SeqScan	NaN	NaN	NaN	NaN
	Sort	NaN	NaN	NaN	NaN
Projection	Aggregation	NaN	NaN	NaN	NaN
	Filter	NaN	NaN	NaN	NaN
	HashJoin	NaN	NaN	NaN	NaN
	NestedLoopCrossJoin	NaN	NaN	NaN	NaN
	NestedLoopJoin	NaN	NaN	NaN	NaN
	SeqScan	NaN	NaN	NaN	NaN
	Sort	NaN	NaN	NaN	NaN
SeqScan	Aggregation	NaN	NaN	NaN	NaN
	Filter	NaN	NaN	NaN	NaN
	HashJoin	NaN	NaN	NaN	NaN
	NestedLoopCrossJoin	NaN	NaN	NaN	NaN
	NestedLoopJoin	NaN	NaN	NaN	NaN
	Projection	NaN	NaN	NaN	NaN
	Sort	NaN	NaN	NaN	NaN
Sort	Aggregation	NaN	NaN	NaN	NaN
	Filter	NaN	NaN	NaN	NaN
	HashJoin	NaN	NaN	NaN	NaN
	NestedLoopCrossJoin	NaN	NaN	NaN	NaN
	NestedLoopJoin	NaN	NaN	NaN	NaN
	Projection	NaN	NaN	NaN	NaN
	SeqScan	NaN	NaN	NaN	NaN

row_count		1024	2048	4096	8192	\
a	b					
Aggregation	Filter	-0.426	-0.222	-0.149	-0.142	
	HashJoin	-0.718	-0.683	-0.713	-0.726	
	NestedLoopCrossJoin	-0.527	NaN	NaN	NaN	
	NestedLoopJoin	-0.304	-0.315	NaN	NaN	
	Projection	-2.116	-1.993	-2.153	-1.999	
	SeqScan	-0.086	-0.019	-0.195	-0.307	
	Sort	-0.619	-0.627	-0.722	-0.719	
Filter	Aggregation	0.426	0.222	0.149	0.142	
	HashJoin	-0.293	-0.461	-0.565	-0.584	
	NestedLoopCrossJoin	-0.101	NaN	NaN	NaN	
	NestedLoopJoin	0.122	-0.094	NaN	NaN	
	Projection	-1.690	-1.771	-2.004	-1.857	
	SeqScan	0.339	0.202	-0.046	-0.165	

	Sort	-0.194	-0.406	-0.574	-0.577
HashJoin	Aggregation	0.718	0.683	0.713	0.726
	Filter	0.293	0.461	0.565	0.584
	NestedLoopCrossJoin	0.192	NaN	NaN	NaN
	NestedLoopJoin	0.415	0.368	NaN	NaN
	Projection	-1.398	-1.310	-1.440	-1.273
	SeqScan	0.632	0.664	0.518	0.419
	Sort	0.099	0.056	-0.009	0.007
NestedLoopCrossJoin	Aggregation	0.527	NaN	NaN	NaN
	Filter	0.101	NaN	NaN	NaN
	HashJoin	-0.192	NaN	NaN	NaN
	NestedLoopJoin	0.223	NaN	NaN	NaN
	Projection	-1.589	NaN	NaN	NaN
	SeqScan	0.441	NaN	NaN	NaN
	Sort	-0.092	NaN	NaN	NaN
NestedLoopJoin	Aggregation	0.304	0.315	NaN	NaN
	Filter	-0.122	0.094	NaN	NaN
	HashJoin	-0.415	-0.368	NaN	NaN
	NestedLoopCrossJoin	-0.223	NaN	NaN	NaN
	Projection	-1.812	-1.678	NaN	NaN
	SeqScan	0.217	0.296	NaN	NaN
	Sort	-0.315	-0.312	NaN	NaN
Projection	Aggregation	2.116	1.993	2.153	1.999
	Filter	1.690	1.771	2.004	1.857
	HashJoin	1.398	1.310	1.440	1.273
	NestedLoopCrossJoin	1.589	NaN	NaN	NaN
	NestedLoopJoin	1.812	1.678	NaN	NaN
	SeqScan	2.030	1.973	1.958	1.691
	Sort	1.497	1.365	1.431	1.280
SeqScan	Aggregation	0.086	0.019	0.195	0.307
	Filter	-0.339	-0.202	0.046	0.165
	HashJoin	-0.632	-0.664	-0.518	-0.419
	NestedLoopCrossJoin	-0.441	NaN	NaN	NaN
	NestedLoopJoin	-0.217	-0.296	NaN	NaN
	Projection	-2.030	-1.973	-1.958	-1.691
	Sort	-0.533	-0.608	-0.528	-0.412
Sort	Aggregation	0.619	0.627	0.722	0.719
	Filter	0.194	0.406	0.574	0.577
	HashJoin	-0.099	-0.056	0.009	-0.007
	NestedLoopCrossJoin	0.092	NaN	NaN	NaN
	NestedLoopJoin	0.315	0.312	NaN	NaN
	Projection	-1.497	-1.365	-1.431	-1.280
	SeqScan	0.533	0.608	0.528	0.412

row_count		16384	32768	65536	131072	\
a	b					
Aggregation	Filter	0.126	0.347	0.440	0.716	

	HashJoin	-0.673	-0.347	-0.063	NaN
	NestedLoopCrossJoin	NaN	NaN	NaN	NaN
	NestedLoopJoin	NaN	NaN	NaN	NaN
	Projection	-1.768	-1.389	-1.083	-0.898
	SeqScan	-0.291	0.146	0.394	0.684
	Sort	-0.647	-0.359	-0.245	-0.307
Filter	Aggregation	-0.126	-0.347	-0.440	-0.716
	HashJoin	-0.799	-0.695	-0.503	NaN
	NestedLoopCrossJoin	NaN	NaN	NaN	NaN
	NestedLoopJoin	NaN	NaN	NaN	NaN
	Projection	-1.894	-1.736	-1.523	-1.614
	SeqScan	-0.417	-0.201	-0.046	-0.031
	Sort	-0.773	-0.706	-0.685	-1.023
HashJoin	Aggregation	0.673	0.347	0.063	NaN
	Filter	0.799	0.695	0.503	NaN
	NestedLoopCrossJoin	NaN	NaN	NaN	NaN
	NestedLoopJoin	NaN	NaN	NaN	NaN
	Projection	-1.095	-1.042	-1.019	NaN
	SeqScan	0.382	0.494	0.458	NaN
	Sort	0.026	-0.011	-0.181	NaN
NestedLoopCrossJoin	Aggregation	NaN	NaN	NaN	NaN
	Filter	NaN	NaN	NaN	NaN
	HashJoin	NaN	NaN	NaN	NaN
	NestedLoopJoin	NaN	NaN	NaN	NaN
	Projection	NaN	NaN	NaN	NaN
	SeqScan	NaN	NaN	NaN	NaN
	Sort	NaN	NaN	NaN	NaN
NestedLoopJoin	Aggregation	NaN	NaN	NaN	NaN
	Filter	NaN	NaN	NaN	NaN
	HashJoin	NaN	NaN	NaN	NaN
	NestedLoopCrossJoin	NaN	NaN	NaN	NaN
	Projection	NaN	NaN	NaN	NaN
	SeqScan	NaN	NaN	NaN	NaN
	Sort	NaN	NaN	NaN	NaN
Projection	Aggregation	1.768	1.389	1.083	0.898
	Filter	1.894	1.736	1.523	1.614
	HashJoin	1.095	1.042	1.019	NaN
	NestedLoopCrossJoin	NaN	NaN	NaN	NaN
	NestedLoopJoin	NaN	NaN	NaN	NaN
	SeqScan	1.477	1.535	1.477	1.582
	Sort	1.121	1.030	0.838	0.591
SeqScan	Aggregation	0.291	-0.146	-0.394	-0.684
	Filter	0.417	0.201	0.046	0.031
	HashJoin	-0.382	-0.494	-0.458	NaN
	NestedLoopCrossJoin	NaN	NaN	NaN	NaN
	NestedLoopJoin	NaN	NaN	NaN	NaN
	Projection	-1.477	-1.535	-1.477	-1.582

	Sort	-0.356	-0.505	-0.639	-0.992
Sort	Aggregation	0.647	0.359	0.245	0.307
	Filter	0.773	0.706	0.685	1.023
	HashJoin	-0.026	0.011	0.181	NaN
	NestedLoopCrossJoin	NaN	NaN	NaN	NaN
	NestedLoopJoin	NaN	NaN	NaN	NaN
	Projection	-1.121	-1.030	-0.838	-0.591
	SeqScan	0.356	0.505	0.639	0.992

row_count		262144
a	b	
Aggregation	Filter	0.910
	HashJoin	NaN
	NestedLoopCrossJoin	NaN
	NestedLoopJoin	NaN
	Projection	-1.023
	SeqScan	0.392
	Sort	-0.407
Filter	Aggregation	-0.910
	HashJoin	NaN
	NestedLoopCrossJoin	NaN
	NestedLoopJoin	NaN
	Projection	-1.933
	SeqScan	-0.518
	Sort	-1.317
HashJoin	Aggregation	NaN
	Filter	NaN
	NestedLoopCrossJoin	NaN
	NestedLoopJoin	NaN
	Projection	NaN
	SeqScan	NaN
	Sort	NaN
NestedLoopCrossJoin	Aggregation	NaN
	Filter	NaN
	HashJoin	NaN
	NestedLoopJoin	NaN
	Projection	NaN
	SeqScan	NaN
	Sort	NaN
NestedLoopJoin	Aggregation	NaN
	Filter	NaN
	HashJoin	NaN
	NestedLoopCrossJoin	NaN
	Projection	NaN
	SeqScan	NaN
	Sort	NaN
Projection	Aggregation	1.023

```
                    Filter                1.933
                    HashJoin                NaN
                    NestedLoopCrossJoin     NaN
                    NestedLoopJoin          NaN
                    SeqScan               1.415
                    Sort                  0.616
SeqScan             Aggregation          -0.392
                    Filter                0.518
                    HashJoin                NaN
                    NestedLoopCrossJoin     NaN
                    NestedLoopJoin          NaN
                    Projection           -1.415
                    Sort                 -0.799
Sort                Aggregation           0.407
                    Filter                1.317
                    HashJoin                NaN
                    NestedLoopCrossJoin     NaN
                    NestedLoopJoin          NaN
                    Projection           -0.616
                    SeqScan               0.799
```

[8]:
```
!cd ~/c/iu9-sql-compiler/ && ./build-release/bin/benchmarks␣
↪--benchmark_filter='OperatorCostMatched*' --benchmark_out=/tmp/
↪operator-cost-matched.json --benchmark_out_format=json
```

```
2026-05-31T16:16:05+03:00
Running ./build-release/bin/benchmarks
Run on (8 X 4200 MHz CPU s)
CPU Caches:
  L1 Data 48 KiB (x4)
  L1 Instruction 32 KiB (x4)
  L2 Unified 1280 KiB (x4)
  L3 Unified 8192 KiB (x1)
Load Average: 3.38, 2.81, 1.77
***WARNING*** CPU scaling is enabled, the benchmark real time measurements may
be noisy and will incur extra overhead.
------------------------------------------------------------------------------
-----------------------------------------------------
Benchmark
Time             CPU   Iterations UserCounters…
------------------------------------------------------------------------------
-----------------------------------------------------
OperatorCostMatched/target_cost:640000/SeqScan/6400/real_time

    532418 ns       521056 ns         1116
model_cost=640k output_rows=6.4k plan_cost=640k rows=6.4k
OperatorCostMatched/target_cost:640000/Filter/6400/real_time

    508270 ns       501445 ns         1089
```

```
model_cost=640k output_rows=3.328k plan_cost=1.28M rows=6.4k
OperatorCostMatched/target_cost:640000/Projection/29091/real_time
   4068856 ns      4010747 ns          156
model_cost=640.002k output_rows=29.091k plan_cost=3.5491M
rows=29.091k
OperatorCostMatched/target_cost:640000/Sort/4476/real_time
    997566 ns       988471 ns          786
model_cost=640.068k output_rows=4.476k plan_cost=1.08767M
rows=4.476k
OperatorCostMatched/target_cost:640000/Aggregation/1255/real_time
    529187 ns       520615 ns         1000
model_cost=640.05k output_rows=1.255k plan_cost=765.55k rows=1.255k
OperatorCostMatched/target_cost:640000/HashJoin/4638/4638/real_time
   1026885 ns      1016820 ns          714
lhs_rows=4.638k model_cost=640.044k output_rows=4.638k
plan_cost=1.56764M rhs_rows=4.638k
OperatorCostMatched/target_cost:640000/NestedLoopJoin/96/96/real_time
    875389 ns       866351 ns          804 lhs_rows=96
model_cost=645.12k output_rows=96 plan_cost=664.32k rhs_rows=96
OperatorCostMatched/target_cost:640000/NestedLoopCrossJoin/78/78/real_
time         527338 ns       522288 ns         1107
lhs_rows=78 model_cost=632.736k output_rows=6.084k
plan_cost=648.336k rhs_rows=78
OperatorCostMatched/target_cost:1000000/SeqScan/10000/real_time
    507202 ns       502389 ns         1000
model_cost=1M output_rows=10k plan_cost=1M rows=10k
OperatorCostMatched/target_cost:1000000/Filter/10000/real_time
    861315 ns       854085 ns          818
model_cost=1M output_rows=5.12k plan_cost=2M rows=10k
OperatorCostMatched/target_cost:1000000/Projection/45455/real_time
   6698145 ns      6632289 ns           94
model_cost=1.00001M output_rows=45.455k plan_cost=5.54551M
rows=45.455k
OperatorCostMatched/target_cost:1000000/Sort/6993/real_time
   1503435 ns      1491454 ns          442
model_cost=0.999999M output_rows=6.993k plan_cost=1.6993M
rows=6.993k
OperatorCostMatched/target_cost:1000000/Aggregation/1961/real_time
    702024 ns       697631 ns          968
model_cost=1.00011M output_rows=1.961k plan_cost=1.19621M
rows=1.961k
```

```
OperatorCostMatched/target_cost:1000000/HashJoin/7246/7246/real_time
   1406982 ns      1398061 ns          445
lhs_rows=7.246k model_cost=0.999948M output_rows=7.246k
plan_cost=2.44915M rhs_rows=7.246k
OperatorCostMatched/target_cost:1000000/NestedLoopJoin/120/120/real_ti
me          1204952 ns      1197240 ns          548
lhs_rows=120 model_cost=1.008M output_rows=120 plan_cost=1.032M
rhs_rows=120
OperatorCostMatched/target_cost:1000000/NestedLoopCrossJoin/98/98/real
_time        735577 ns       729661 ns          851
lhs_rows=98 model_cost=998.816k output_rows=9.604k
plan_cost=1.01842M rhs_rows=98
OperatorCostMatched/target_cost:3240000/SeqScan/32400/real_time
   1552618 ns      1542405 ns          398
model_cost=3.24M output_rows=32.4k plan_cost=3.24M rows=32.4k
OperatorCostMatched/target_cost:3240000/Filter/32400/real_time
   2615187 ns      2600696 ns          253
model_cost=3.24M output_rows=16.384k plan_cost=6.48M rows=32.4k
OperatorCostMatched/target_cost:3240000/Projection/147273/real_time
  24615070 ns     24428312 ns           28
model_cost=3.24001M output_rows=147.273k plan_cost=17.9673M
rows=147.273k
OperatorCostMatched/target_cost:3240000/Sort/19636/real_time
   5716537 ns      5678245 ns          122
model_cost=3.23994M output_rows=19.636k plan_cost=5.20354M
rows=19.636k
OperatorCostMatched/target_cost:3240000/Aggregation/6353/real_time
   2528047 ns      2507421 ns          253
model_cost=3.24003M output_rows=6.353k plan_cost=3.87533M
rows=6.353k
OperatorCostMatched/target_cost:3240000/HashJoin/23478/23478/real_time
   5300551 ns      5262456 ns          116
lhs_rows=23.478k model_cost=3.23996M output_rows=23.478k
plan_cost=7.93556M rhs_rows=23.478k
OperatorCostMatched/target_cost:3240000/NestedLoopJoin/215/215/real_ti
me          3440769 ns      3423785 ns          199
lhs_rows=215 model_cost=3.23575M output_rows=215 plan_cost=3.27875M
rhs_rows=215
OperatorCostMatched/target_cost:3240000/NestedLoopCrossJoin/177/177/re
al_time     2294554 ns      2274793 ns          241
lhs_rows=177 model_cost=3.25822M output_rows=31.329k
```

```
plan_cost=3.29362M rhs_rows=177
OperatorCostMatched/target_cost:6760000/SeqScan/67600/real_time
   4220053 ns      4185107 ns          169
model_cost=6.76M output_rows=67.6k plan_cost=6.76M rows=67.6k
OperatorCostMatched/target_cost:6760000/Filter/67600/real_time
   5592034 ns      5560841 ns          126
model_cost=6.76M output_rows=33.808k plan_cost=13.52M rows=67.6k
OperatorCostMatched/target_cost:6760000/Projection/307273/real_time
  51498743 ns     51127846 ns           14
model_cost=6.76001M output_rows=307.273k plan_cost=37.4873M
rows=307.273k
OperatorCostMatched/target_cost:6760000/Sort/38409/real_time
  12802465 ns     12732765 ns           50
model_cost=6.75998M output_rows=38.409k plan_cost=10.6009M
rows=38.409k
OperatorCostMatched/target_cost:6760000/Aggregation/13255/real_time
   5687200 ns      5655535 ns          118
model_cost=6.76005M output_rows=13.255k plan_cost=8.08555M
rows=13.255k
OperatorCostMatched/target_cost:6760000/HashJoin/48986/48986/real_time
  13564179 ns     13440945 ns           51
lhs_rows=48.986k model_cost=6.76007M output_rows=48.986k
plan_cost=16.5573M rhs_rows=48.986k
OperatorCostMatched/target_cost:6760000/NestedLoopJoin/311/311/real_ti
me          7198310 ns      7163077 ns           93
lhs_rows=311 model_cost=6.77047M output_rows=311 plan_cost=6.83267M
rhs_rows=311
OperatorCostMatched/target_cost:6760000/NestedLoopCrossJoin/255/255/re
al_time     6012240 ns      5936779 ns           87
lhs_rows=255 model_cost=6.7626M output_rows=65.025k
plan_cost=6.8136M rhs_rows=255
OperatorCostMatched/target_cost:12250000/SeqScan/122500/real_time
   7610143 ns      7525813 ns           83
model_cost=12.25M output_rows=122.5k plan_cost=12.25M rows=122.5k
OperatorCostMatched/target_cost:12250000/Filter/122500/real_time
  10313976 ns     10227032 ns           63
model_cost=12.25M output_rows=61.44k plan_cost=24.5M rows=122.5k
OperatorCostMatched/target_cost:12250000/Projection/556818/real_time
  90203371 ns     89533850 ns            8
model_cost=12.25M output_rows=556.818k plan_cost=67.9318M
rows=556.818k
```

```
OperatorCostMatched/target_cost:12250000/Sort/65536/real_time
  26495662 ns     26260681 ns           22
model_cost=12.2552M output_rows=65.536k plan_cost=18.8088M
rows=65.536k
OperatorCostMatched/target_cost:12250000/Aggregation/24020/real_time
  15047581 ns     14860662 ns           50
model_cost=12.2502M output_rows=24.02k plan_cost=14.6522M
rows=24.02k
OperatorCostMatched/target_cost:12250000/HashJoin/88768/88768/real_tim
e          25706484 ns     25401595 ns           28
lhs_rows=88.768k model_cost=12.25M output_rows=88.768k
plan_cost=30.0036M rhs_rows=88.768k
OperatorCostMatched/target_cost:12250000/NestedLoopJoin/418/418/real_t
ime        12686905 ns     12628552 ns           47
lhs_rows=418 model_cost=12.2307M output_rows=418 plan_cost=12.3143M
rhs_rows=418
OperatorCostMatched/target_cost:12250000/NestedLoopCrossJoin/343/343/r
eal_time   12634892 ns     12474012 ns           52
lhs_rows=343 model_cost=12.2355M output_rows=117.649k
plan_cost=12.3041M rhs_rows=343
OperatorCostMatched/target_cost:26010000/SeqScan/260100/real_time
  18734465 ns     18479865 ns           37
model_cost=26.01M output_rows=260.1k plan_cost=26.01M rows=260.1k
OperatorCostMatched/target_cost:26010000/Filter/260100/real_time
  26934394 ns     26755899 ns           28
model_cost=26.01M output_rows=130.052k plan_cost=52.02M rows=260.1k
OperatorCostMatched/target_cost:26010000/Projection/1182273/real_time
 202987386 ns    201290217 ns            4
model_cost=26.01M output_rows=1.18227M plan_cost=144.237M
rows=1.18227M
OperatorCostMatched/target_cost:26010000/Sort/131364/real_time
  74202993 ns     73103763 ns            9
model_cost=26.0101M output_rows=131.364k plan_cost=39.1465M
rows=131.364k
OperatorCostMatched/target_cost:26010000/Aggregation/51000/real_time
  48595095 ns     48139275 ns           16
model_cost=26.01M output_rows=51k plan_cost=31.11M rows=51k
OperatorCostMatched/target_cost:26010000/HashJoin/188478/188478/real_t
ime        52584297 ns     52158890 ns           14
lhs_rows=188.478k model_cost=26.01M output_rows=188.478k
plan_cost=63.7056M rhs_rows=188.478k
```

```
OperatorCostMatched/target_cost:26010000/NestedLoopJoin/610/610/real_t
ime        25448562 ns     25343677 ns           26
lhs_rows=610 model_cost=26.047M output_rows=610 plan_cost=26.169M
rhs_rows=610
OperatorCostMatched/target_cost:26010000/NestedLoopCrossJoin/500/500/r
eal_time   30993358 ns     30600644 ns           27
lhs_rows=500 model_cost=26M output_rows=250k plan_cost=26.1M
rhs_rows=500
```

[ ]:
```
import json
results = None
with open('/tmp/operator-cost-matched.json') as f:
    results = json.load(f)

results
```

[10]:
```
matched_df = pd.DataFrame(results['benchmarks'])
matched_df = matched_df[matched_df['run_type'] == 'iteration'].copy()

parts = matched_df['name'].str.split('/')
matched_df['target_cost'] = parts.str[1].str.removeprefix('target_cost:').
 ↪astype(int)
matched_df['operator'] = parts.str[2]
matched_df['left_rows'] = parts.str[3].astype(int)
matched_df['right_rows'] = parts.apply(lambda p: int(p[4]) if len(p) == 6 else␣
 ↪None)
matched_df['real_time_ms'] = matched_df['real_time'] / 1e6

matched_df['matching_error'] = (matched_df['model_cost'] -␣
 ↪matched_df['target_cost']).abs() / matched_df['target_cost']
assert (matched_df['matching_error'] < 0.02).all()

matched_df[['target_cost', 'operator', 'left_rows', 'right_rows',␣
 ↪'real_time_ms', 'model_cost', 'plan_cost', 'matching_error']].
 ↪sort_values(['target_cost', 'operator'])
```

[10]:

	target_cost	operator	left_rows	right_rows	real_time_ms	\
4	640000	Aggregation	1255	NaN	0.529187	
1	640000	Filter	6400	NaN	0.508270	
5	640000	HashJoin	4638	4638.0	1.026885	
7	640000	NestedLoopCrossJoin	78	78.0	0.527338	
6	640000	NestedLoopJoin	96	96.0	0.875389	
2	640000	Projection	29091	NaN	4.068856	
0	640000	SeqScan	6400	NaN	0.532418	
3	640000	Sort	4476	NaN	0.997566	

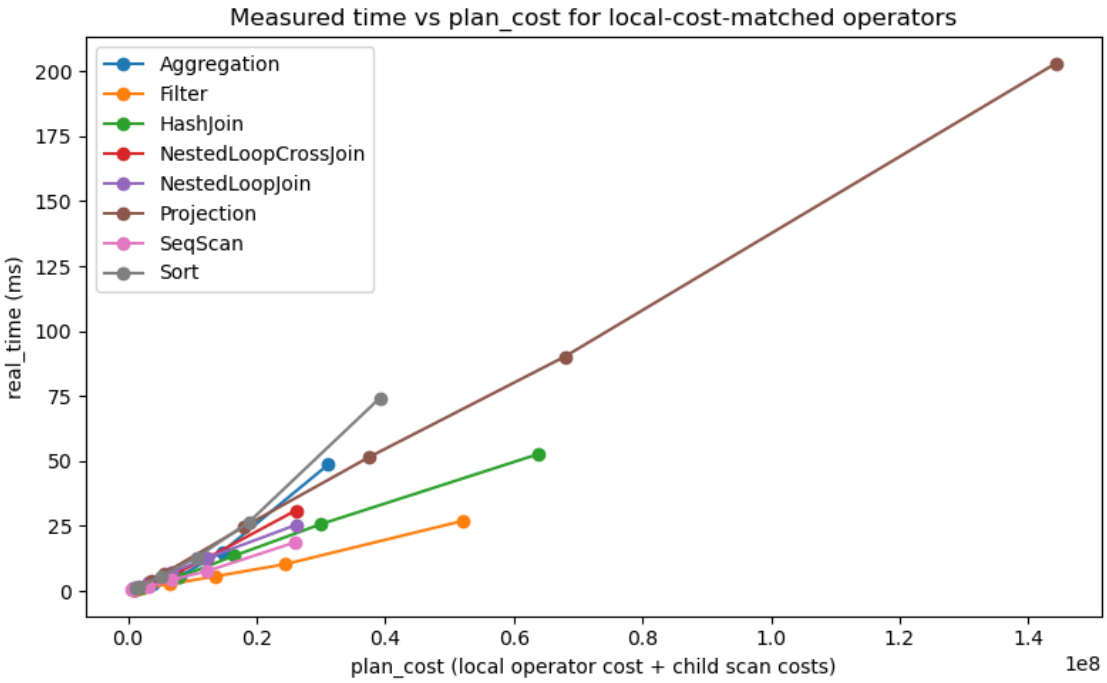
12	1000000	Aggregation	1961	NaN	0.702024
9	1000000	Filter	10000	NaN	0.861315
13	1000000	HashJoin	7246	7246.0	1.406982
15	1000000	NestedLoopCrossJoin	98	98.0	0.735577
14	1000000	NestedLoopJoin	120	120.0	1.204952
10	1000000	Projection	45455	NaN	6.698145
8	1000000	SeqScan	10000	NaN	0.507202
11	1000000	Sort	6993	NaN	1.503435
20	3240000	Aggregation	6353	NaN	2.528047
17	3240000	Filter	32400	NaN	2.615187
21	3240000	HashJoin	23478	23478.0	5.300551
23	3240000	NestedLoopCrossJoin	177	177.0	2.294554
22	3240000	NestedLoopJoin	215	215.0	3.440769
18	3240000	Projection	147273	NaN	24.615070
16	3240000	SeqScan	32400	NaN	1.552618
19	3240000	Sort	19636	NaN	5.716537
28	6760000	Aggregation	13255	NaN	5.687200
25	6760000	Filter	67600	NaN	5.592034
29	6760000	HashJoin	48986	48986.0	13.564179
31	6760000	NestedLoopCrossJoin	255	255.0	6.012240
30	6760000	NestedLoopJoin	311	311.0	7.198310
26	6760000	Projection	307273	NaN	51.498743
24	6760000	SeqScan	67600	NaN	4.220053
27	6760000	Sort	38409	NaN	12.802465
36	12250000	Aggregation	24020	NaN	15.047581
33	12250000	Filter	122500	NaN	10.313976
37	12250000	HashJoin	88768	88768.0	25.706484
39	12250000	NestedLoopCrossJoin	343	343.0	12.634892
38	12250000	NestedLoopJoin	418	418.0	12.686905
34	12250000	Projection	556818	NaN	90.203371
32	12250000	SeqScan	122500	NaN	7.610143
35	12250000	Sort	65536	NaN	26.495662
44	26010000	Aggregation	51000	NaN	48.595095
41	26010000	Filter	260100	NaN	26.934394
45	26010000	HashJoin	188478	188478.0	52.584297
47	26010000	NestedLoopCrossJoin	500	500.0	30.993358
46	26010000	NestedLoopJoin	610	610.0	25.448562
42	26010000	Projection	1182273	NaN	202.987386
40	26010000	SeqScan	260100	NaN	18.734465
43	26010000	Sort	131364	NaN	74.202993

	model_cost	plan_cost	matching_error
4	640050.0	765550.0	7.812500e-05
1	640000.0	1280000.0	0.000000e+00
5	640044.0	1567644.0	6.875000e-05
7	632736.0	648336.0	1.135000e-02
6	645120.0	664320.0	8.000000e-03

```
2     640002.0    3549102.0    3.125000e-06
0     640000.0     640000.0    0.000000e+00
3     640068.0    1087668.0    1.062500e-04
12   1000110.0    1196210.0    1.100000e-04
9    1000000.0    2000000.0    0.000000e+00
13    999948.0    2449148.0    5.200000e-05
15    998816.0    1018416.0    1.184000e-03
14   1008000.0    1032000.0    8.000000e-03
10   1000010.0    5545510.0    1.000000e-05
8    1000000.0    1000000.0    0.000000e+00
11    999999.0    1699299.0    1.000000e-06
20   3240030.0    3875330.0    9.259259e-06
17   3240000.0    6480000.0    0.000000e+00
21   3239964.0    7935564.0    1.111111e-05
23   3258216.0    3293616.0    5.622222e-03
22   3235750.0    3278750.0    1.311728e-03
18   3240006.0   17967306.0    1.851852e-06
16   3240000.0    3240000.0    0.000000e+00
19   3239940.0    5203540.0    1.851852e-05
28   6760050.0    8085550.0    7.396450e-06
25   6760000.0   13520000.0    0.000000e+00
29   6760068.0   16557268.0    1.005917e-05
31   6762600.0    6813600.0    3.846154e-04
30   6770470.0    6832670.0    1.548817e-03
26   6760006.0   37487306.0    8.875740e-07
24   6760000.0    6760000.0    0.000000e+00
27   6759984.0   10600884.0    2.366864e-06
36  12250200.0   14652200.0    1.632653e-05
33  12250000.0   24500000.0    0.000000e+00
37  12249984.0   30003584.0    1.306122e-06
39  12235496.0   12304096.0    1.184000e-03
38  12230680.0   12314280.0    1.577143e-03
34  12249996.0   67931796.0    3.265306e-07
32  12250000.0   12250000.0    0.000000e+00
35  12255232.0   18808832.0    4.271020e-04
44  26010000.0   31110000.0    0.000000e+00
41  26010000.0   52020000.0    0.000000e+00
45  26009964.0   63705564.0    1.384083e-06
47  26000000.0   26100000.0    3.844675e-04
46  26047000.0   26169000.0    1.422530e-03
42  26010006.0  144237306.0    2.306805e-07
40  26010000.0   26010000.0    0.000000e+00
43  26010072.0   39146472.0    2.768166e-06
```

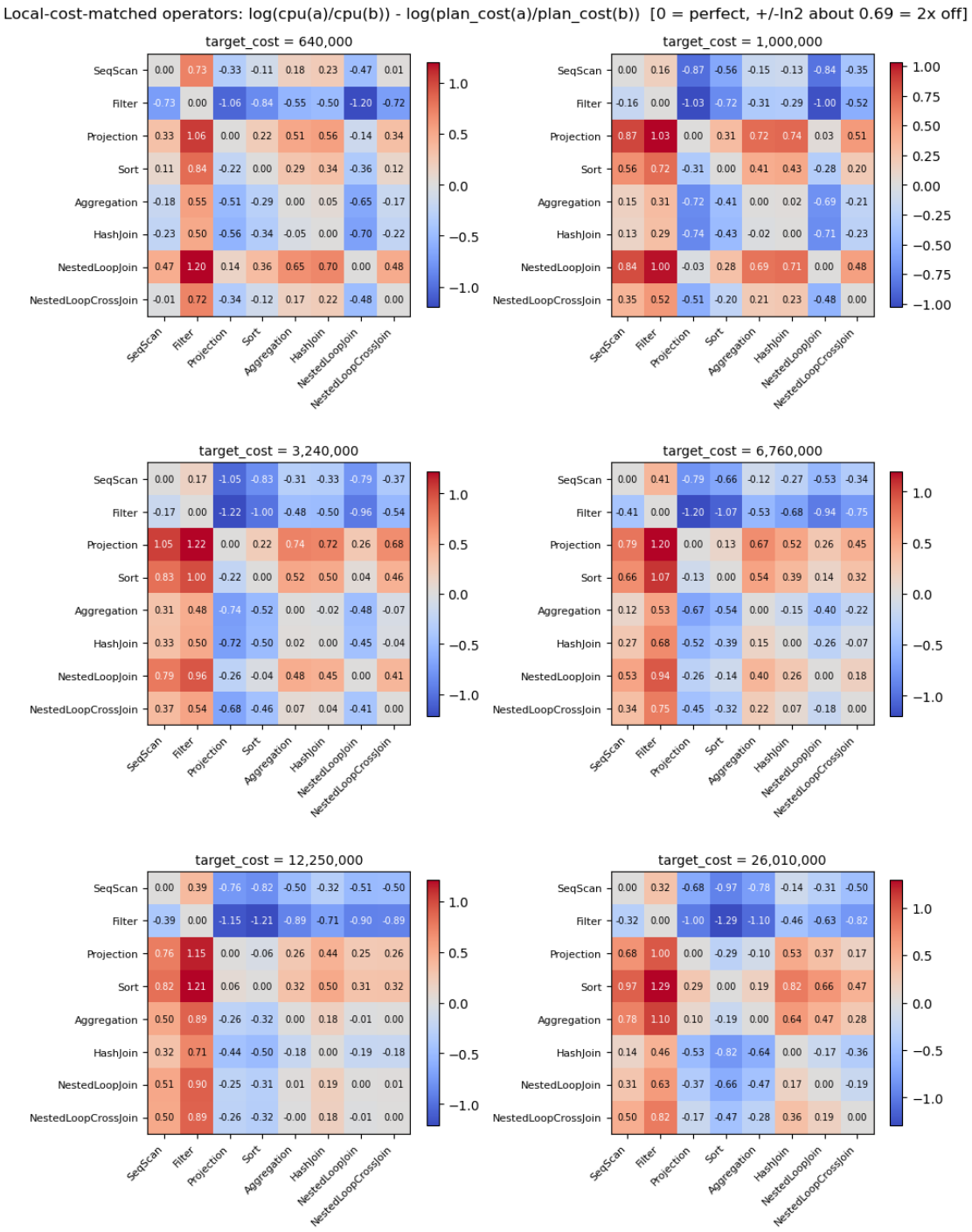
[11]:
```
fig, ax = plt.subplots(figsize=(8, 5))
for op, grp in matched_df.groupby('operator'):
    grp = grp.sort_values('plan_cost')
```

```
    ax.plot(grp['plan_cost'], grp['real_time_ms'], marker='o', label=op)

ax.set_xlabel('plan_cost (local operator cost + child scan costs)')
ax.set_ylabel('real_time (ms)')
ax.set_title('Measured time vs plan_cost for local-cost-matched operators')
ax.legend()
plt.tight_layout()
plt.show()
```



[12]:
```
matched_pivot = matched_df.set_index(['target_cost', 'operator'])[['cpu_time',␣
 ↪'model_cost', 'plan_cost']]
matched_costs = sorted(matched_df['target_cost'].unique())

ncols = 2
nrows = (len(matched_costs) + ncols - 1) // ncols
fig, axes = plt.subplots(nrows, ncols, figsize=(5 * ncols, 4.5 * nrows),␣
 ↪squeeze=False)

for idx, target_cost in enumerate(matched_costs):
    ax = axes[idx // ncols][idx % ncols]
    sub = matched_pivot.xs(target_cost, level='target_cost')
    ops = sub.index.tolist()
    m = len(ops)
```

```
    mat = np.zeros((m, m))
    for i, a in enumerate(ops):
        for j, b in enumerate(ops):
            if i != j:
                mat[i, j] = (np.log(sub.loc[a, 'cpu_time']) - np.log(sub.loc[b,␣
 ↪'cpu_time']) -
                             (np.log(sub.loc[a, 'plan_cost']) - np.log(sub.
 ↪loc[b, 'plan_cost'])))

    vmax = np.abs(mat).max() or 1.0
    im = ax.imshow(mat, cmap='coolwarm', vmin=-vmax, vmax=vmax)
    ax.set_xticks(range(m)); ax.set_xticklabels(ops, rotation=45, ha='right',␣
 ↪fontsize=8)
    ax.set_yticks(range(m)); ax.set_yticklabels(ops, fontsize=8)
    ax.set_title(f'target_cost = {target_cost:,}', fontsize=10)
    plt.colorbar(im, ax=ax, shrink=0.8)
    for i in range(m):
        for j in range(m):
            ax.text(j, i, f'{mat[i,j]:.2f}', ha='center', va='center',␣
 ↪fontsize=7,
                    color='black' if abs(mat[i,j]) < vmax * 0.6 else 'white')

for idx in range(len(matched_costs), nrows * ncols):
    axes[idx // ncols][idx % ncols].set_visible(False)

fig.suptitle('Local-cost-matched operators: log(cpu(a)/cpu(b)) -␣
 ↪log(plan_cost(a)/plan_cost(b))  [0 = perfect, +/-ln2 about 0.69 = 2x off]',␣
 ↪fontsize=12)
plt.tight_layout()
plt.show()
```



[13]:
```
matched_rows = []
for target_cost in matched_costs:
    sub = matched_pivot.xs(target_cost, level='target_cost')
    ops = sub.index.tolist()
```

```
    for a in ops:
        for b in ops:
            if a != b:
                delta = (np.log(sub.loc[a, 'cpu_time']) - np.log(sub.loc[b,␣
 ↪'cpu_time']) -
                         (np.log(sub.loc[a, 'plan_cost']) - np.log(sub.loc[b,␣
 ↪'plan_cost'])))
                matched_rows.append({'target_cost': target_cost, 'a': a, 'b':␣
 ↪b, 'delta': delta})

matched_comparison = pd.DataFrame(matched_rows)
matched_comparison.pivot_table(index=['a', 'b'], columns='target_cost',␣
 ↪values='delta').round(3)
```

[13]:

target_cost		640000	1000000	3240000
a	b			
Aggregation	Filter	0.552	0.312	0.478
	HashJoin	0.047	0.021	-0.025
	NestedLoopCrossJoin	-0.169	-0.206	-0.065
	NestedLoopJoin	-0.651	-0.688	-0.479
	Projection	-0.508	-0.718	-0.743
	SeqScan	-0.180	0.149	0.307
	Sort	-0.290	-0.409	-0.523
Filter	Aggregation	-0.552	-0.312	-0.478
	HashJoin	-0.504	-0.290	-0.502
	NestedLoopCrossJoin	-0.721	-0.517	-0.543
	NestedLoopJoin	-1.203	-0.999	-0.956
	Projection	-1.059	-1.030	-1.220
	SeqScan	-0.732	-0.162	-0.171
	Sort	-0.841	-0.720	-1.000
HashJoin	Aggregation	-0.047	-0.021	0.025
	Filter	0.504	0.290	0.502
	NestedLoopCrossJoin	-0.217	-0.227	-0.041
	NestedLoopJoin	-0.698	-0.709	-0.454
	Projection	-0.555	-0.740	-0.718
	SeqScan	-0.227	0.128	0.331
	Sort	-0.337	-0.430	-0.498
NestedLoopCrossJoin	Aggregation	0.169	0.206	0.065
	Filter	0.721	0.517	0.543
	HashJoin	0.217	0.227	0.041
	NestedLoopJoin	-0.482	-0.482	-0.413
	Projection	-0.338	-0.512	-0.677
	SeqScan	-0.011	0.355	0.372
	Sort	-0.121	-0.203	-0.457
NestedLoopJoin	Aggregation	0.651	0.688	0.479
	Filter	1.203	0.999	0.956
	HashJoin	0.698	0.709	0.454

	NestedLoopCrossJoin	0.482	0.482	0.413
	Projection	0.143	-0.030	-0.264
	SeqScan	0.471	0.837	0.786
	Sort	0.361	0.279	-0.044
Projection	Aggregation	0.508	0.718	0.743
	Filter	1.059	1.030	1.220
	HashJoin	0.555	0.740	0.718
	NestedLoopCrossJoin	0.338	0.512	0.677
	NestedLoopJoin	-0.143	0.030	0.264
	SeqScan	0.328	0.867	1.049
	Sort	0.218	0.309	0.220
SeqScan	Aggregation	0.180	-0.149	-0.307
	Filter	0.732	0.162	0.171
	HashJoin	0.227	-0.128	-0.331
	NestedLoopCrossJoin	0.011	-0.355	-0.372
	NestedLoopJoin	-0.471	-0.837	-0.786
	Projection	-0.328	-0.867	-1.049
	Sort	-0.110	-0.558	-0.830
Sort	Aggregation	0.290	0.409	0.523
	Filter	0.841	0.720	1.000
	HashJoin	0.337	0.430	0.498
	NestedLoopCrossJoin	0.121	0.203	0.457
	NestedLoopJoin	-0.361	-0.279	0.044
	Projection	-0.218	-0.309	-0.220
	SeqScan	0.110	0.558	0.830

target_cost		6760000	12250000	26010000
a	b			
Aggregation	Filter	0.531	0.888	1.101
	HashJoin	-0.149	0.181	0.637
	NestedLoopCrossJoin	-0.220	0.000	0.277
	NestedLoopJoin	-0.405	-0.011	0.469
	Projection	-0.668	-0.262	0.103
	SeqScan	0.122	0.501	0.778
	Sort	-0.541	-0.320	-0.188
Filter	Aggregation	-0.531	-0.888	-1.101
	HashJoin	-0.680	-0.707	-0.465
	NestedLoopCrossJoin	-0.751	-0.887	-0.824
	NestedLoopJoin	-0.936	-0.899	-0.633
	Projection	-1.199	-1.150	-0.998
	SeqScan	-0.409	-0.386	-0.323
	Sort	-1.072	-1.207	-1.289
HashJoin	Aggregation	0.149	-0.181	-0.637
	Filter	0.680	0.707	0.465
	NestedLoopCrossJoin	-0.071	-0.180	-0.359
	NestedLoopJoin	-0.256	-0.192	-0.168
	Projection	-0.519	-0.443	-0.533

	SeqScan	0.271	0.321	0.142
	Sort	-0.392	-0.500	-0.825
NestedLoopCrossJoin	Aggregation	0.220	-0.000	-0.277
	Filter	0.751	0.887	0.824
	HashJoin	0.071	0.180	0.359
	NestedLoopJoin	-0.185	-0.011	0.191
	Projection	-0.448	-0.262	-0.174
	SeqScan	0.342	0.501	0.501
	Sort	-0.321	-0.320	-0.465
NestedLoopJoin	Aggregation	0.405	0.011	-0.469
	Filter	0.936	0.899	0.633
	HashJoin	0.256	0.192	0.168
	NestedLoopCrossJoin	0.185	0.011	-0.191
	Projection	-0.263	-0.251	-0.365
	SeqScan	0.527	0.512	0.310
	Sort	-0.136	-0.309	-0.657
Projection	Aggregation	0.668	0.262	-0.103
	Filter	1.199	1.150	0.998
	HashJoin	0.519	0.443	0.533
	NestedLoopCrossJoin	0.448	0.262	0.174
	NestedLoopJoin	0.263	0.251	0.365
	SeqScan	0.790	0.763	0.675
	Sort	0.127	-0.058	-0.291
SeqScan	Aggregation	-0.122	-0.501	-0.778
	Filter	0.409	0.386	0.323
	HashJoin	-0.271	-0.321	-0.142
	NestedLoopCrossJoin	-0.342	-0.501	-0.501
	NestedLoopJoin	-0.527	-0.512	-0.310
	Projection	-0.790	-0.763	-0.675
	Sort	-0.663	-0.821	-0.966
Sort	Aggregation	0.541	0.320	0.188
	Filter	1.072	1.207	1.289
	HashJoin	0.392	0.500	0.825
	NestedLoopCrossJoin	0.321	0.320	0.465
	NestedLoopJoin	0.136	0.309	0.657
	Projection	-0.127	0.058	0.291
	SeqScan	0.663	0.821	0.966